



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
01-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TIAS, S, I(A), I, PHASE LIVE-I & I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A gas undergoes a process in which pressure of gas is P and volume of gas V are related as $VP^n = \text{constant}$ then choose the correct statement(s) :

(A) Bulk modulus of gas is $\frac{P}{n}$.

(B) Volume expansion coefficient is $\frac{n}{(n-1)T}$.

(C) Work done in process is $\int dW = -n \int V dP$

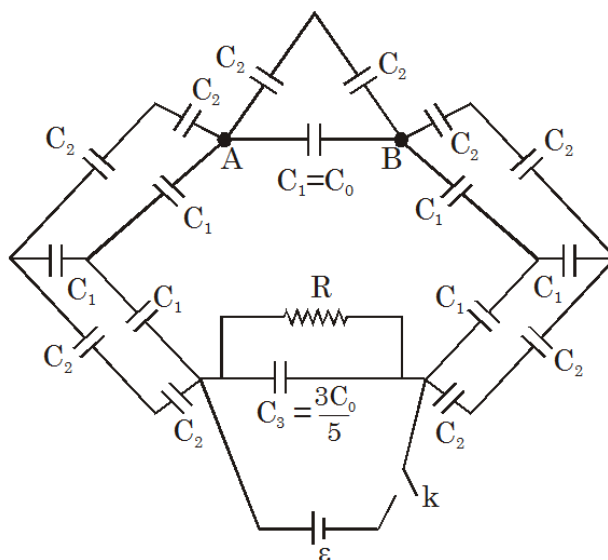
(D) None

Space for Rough Work

2. A solid metal sphere has a charge $+3Q$ and radius is a which is concentric with conducting spherical shell of radius b having charge $-Q$. Choose the correct statement(s) :

- (A) Electric field at distance R ($a < R < b$) is $\frac{1}{4\pi\epsilon_0} \frac{3Q}{R^2}$
- (B) Electric field at distance R ($R > b$) is $\frac{1}{4\pi\epsilon_0} \left(\frac{2Q}{R^2} \right)$
- (C) Electric potential at R ($a < R < b$) is $\frac{1}{4\pi\epsilon_0} \left(\frac{3Q}{R} - \frac{Q}{b} \right)$
- (D) Electric potential at R ($R > b$) is $\frac{1}{4\pi\epsilon_0} \left(\frac{2Q}{R} \right)$

3. A circuit diagram is shown with an ideal DC battery of emf ϵ , resistance (R) and capacitors having capacitance either C_1 or C_2 or C_3 as shown in circuit diagram, relation between C_1 , C_2 and C_3 is $C_1 = \frac{C_2}{2} = \frac{5C_3}{3} = C_0$. If key is closed at $t = 0$ and initially all the capacitors are uncharged, then choose the correct statement(s) :



- (A) Current in the resistance R remains constant after key is closed.
- (B) Time constant for equivalent R - C circuit is $\frac{1}{RC_0}$
- (C) Potential difference across point A & B is $\frac{\epsilon}{5}$.
- (D) Charge on C_3 is equal to $\frac{3C_0\epsilon}{5}$ if key is closed for infinite time.

Space for Rough Work

4. A satellite is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to mass of earth then choose the correct statement(s) :
- (A) The acceleration of satellite is always directed towards earth.
 (B) Angular momentum of satellite about the centre of earth remains conserved.
 (C) Kinetic energy of satellite varies periodically with time.
 (D) Total mechanical energy remains conserved.
5. If a uniform conductor of length (ℓ), cross section area (A) and resistivity (ρ) is connected with an ideal DC battery of emf ε then choose the correct option(s) : (Assume n to be no. of e^- per unit volume)
- (A) Current in conductor is $\frac{\varepsilon A}{\rho \ell}$
 (B) Electric field produced in conductor is $\frac{\varepsilon}{\ell}$
 (C) Drift velocity of e^- s in conductor is $\frac{\varepsilon}{\rho \ell n}$
 (D) none
6. An infinite length of cylinder of radius R has uniform charge density ρ then choose the correct option(s) : ($r \rightarrow$ radial distance from central axis of cylinder)
- (A) Electric field at distance r ($r < R$) is proportional to r^1 .
 (B) Electric field at distance r ($r > R$) is proportional to r^{-1} .
 (C) Electric field inside the cylinder is constant.
 (D) Electric field outside the cylinder is constant.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

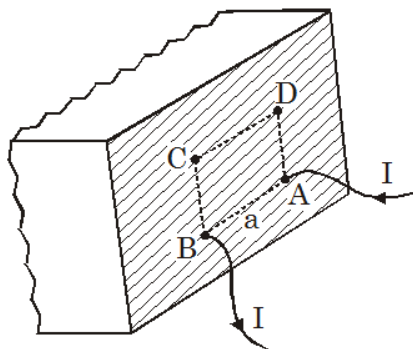
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A plane divides space into two halves. One half is filled with a homogeneous conducting medium and a physicist works in the other. He marks the outline of a square of side a on the plane and let a current I in and out at two of its neighbouring corners A and B using fine electrodes. The potential difference between the other two corners is found to be V as shown in figure. Using superposition principle one can find resistivity ρ of the conducting medium.



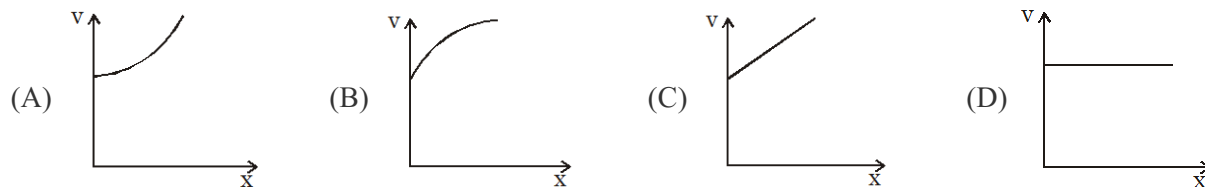
7. The potential difference between points D and C is
- (A) $\frac{I\rho}{2\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)$ (B) $\frac{I\rho}{\pi a}$
- (C) $\frac{I\rho}{\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)$ (D) Zero
8. If r be internal resistance of the battery then the emf of the battery which is supplying current into the wires will be :
- (A) $\frac{I\rho}{2\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) + Ir$ (B) $\frac{I\rho}{\pi a} + Ir$
- (C) $\frac{I\rho}{\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) + Ir$ (D) $\frac{I\rho}{2\pi a} \left(\frac{\sqrt{2}-1}{\sqrt{2}} \right) - Ir$

Space for Rough Work

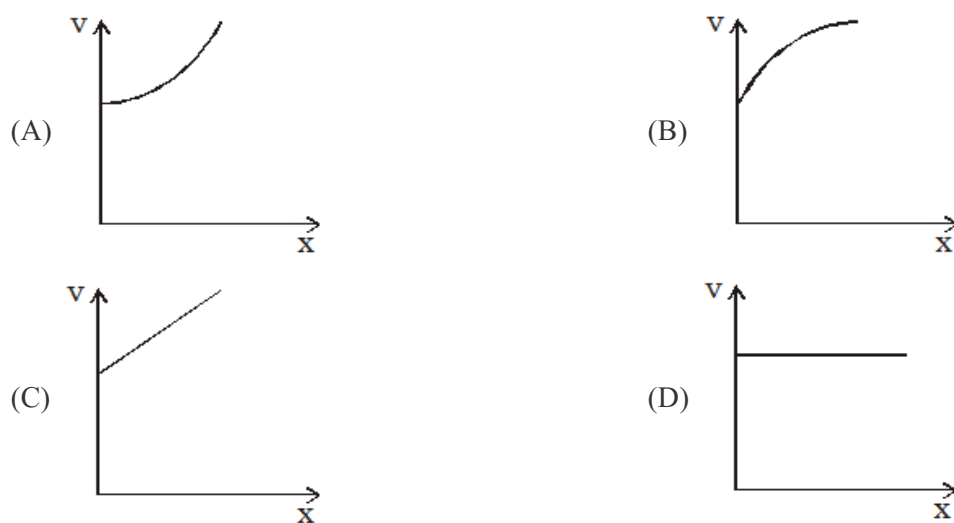
Paragraph for Questions 9 and 10

Between the plates of a parallel plate capacitor a dielectric plate is introduced just to fill the space between the plates. The capacitor is charged and later disconnected from the battery. The dielectric plate is slowly drawn out of the capacitor parallel to the plates.

9. The plot of the potential difference across the plates and the length of the dielectric plate drawn out



10. If the battery is not disconnected and plate is taken out then what would be plot of potential difference and length of dielectric plate taken out.



Space for Rough Work

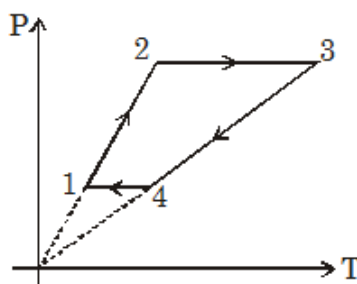
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

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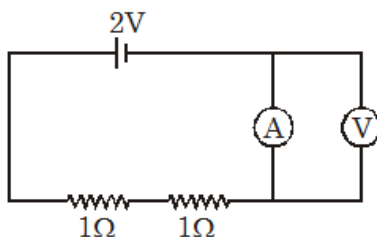
- Two cells E_1 and E_2 when connected in series are balanced on 8m on potentiometer. If the cells are connected with polarities of one of the cell is reversed, they balance on 2m. The ratio of emf's of two cell is $\frac{E_1}{E_2}$ (E_1 is higher) then value of $\frac{E_1}{E_2}$ is.
- Three moles of an ideal monoatomic gas perform a cycle shown in figure. The gas temperature in different states are $T_1 = 200$ K, $T_2 = 400$ K, $T_3 = 1600$ K and $T_4 = 800$ K. The work done (in kJ) by the gas during the cycle is (Take : $R = \frac{25}{3}$ J/mol-K)



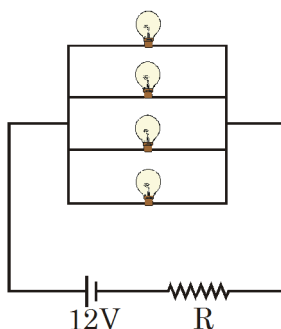
- There are two conducting spheres A & B of equal radius. A is uncharge and B is given charge Q_0 . Now both are touched for some time and then seperated away, now again charge on B made Q_0 and both are touched for some time and seperated away and it is done for n times including first touch and making charge on B is Q_0 again then after n touch charge on sphere A becomes $\left(\frac{x^n - 1}{y^n}\right) Q_0$ then value of $\frac{x}{y}$ is.

Space for Rough Work

4. An air bubble in sphere having 4 cm diameter appears 1cm from nearest to eye when looked along diameter. If inside sphere $\mu = 1.5$, then the distance (in cm) of bubble from refracting surface is
5. In the circuit shown ammeter and voltmeter are ideal and reading of voltmeter is V volts and reading of ammeter is i Amp then $\frac{V}{i}$ is :



6. Four identical electric lamps are labelled 1.5 V, 0.5 A which describes the condition necessary for them to operate at normal brightness. A 12 V battery of negligible internal resistance is connected to lamps as shown then the value of R (in Ω) for normal brightness of each lamp is:



Space for Rough Work

7. An electric dipole is placed along x-axis at origin O. A point P is at a distance of 20 cm from origin. Such that OP makes an angle $\frac{\pi}{3}$ with x-axis. If the electric field at P make an angle $\frac{\pi}{3} + \tan^{-1}(n)$ with x-axis then the value of n is :
8. There are two vessels, each of them contains one mole of a monatomic ideal gas. Initial volume of the gas in each vessel is $8.3 \times 10^{-3} \text{ m}^3$ at 27°C . Equal amount of heat is supplied to the two vessels such that in one of the vessels, the volume of the gas is doubled without change in internal energy, whereas the volume of the gas is held constant in the second vessel. The vessels are now connected to allow free mixing of the gas. The temperature of the combined system (in K) is found to be ($R = 8.3 \text{ J/mol. K}$ and $\log_e 2 = 0.693$)

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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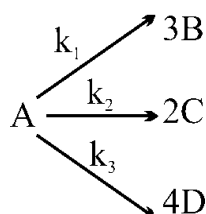
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For the I order parallel reaction



If $k_1 : k_2 : k_3$ is 1 : 2 : 3, then, select the correct statement(s) ($[A]_0 = 1 \text{ M}$)

(A) at ∞ time $[C]$ is $2/3 \text{ M}$

(B) $[B]_t > [C]_t$

(C) When $[A]_t = 1/2 \text{ M}$ then $[D] = 1 \text{ M}$

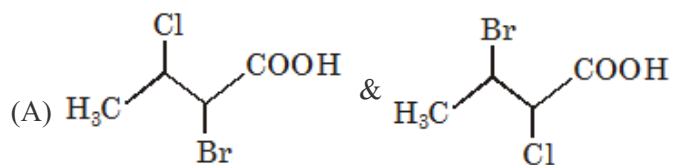
(D) $[A]_t + \frac{[B]_t}{3} + \frac{[C]_t}{2} + \frac{[D]_t}{4} = 1 \text{ M}$

Space for Rough Work

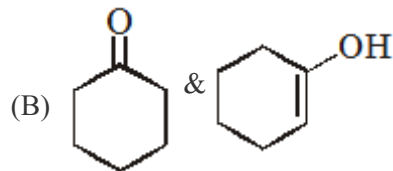
2. Which of the following reaction is / are **INCORRECT** about metallurgical process of copper metal?
- (A) $\text{CuFeS}_2 + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{CuO} + \text{SO}_2$ (B) $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$
- (C) $\text{CuO} + \text{SiO}_2 \rightarrow \text{CuSiO}_3$ (D) $\text{FeO} + \text{SiO}_2 \longrightarrow \text{FeSiO}_3$
3. Which of the following statement(s) is/are **CORRECT** related to extraction of "Tin" from cassiterite (tinstone)
- (A) Iron tungstate is removed by magnetic separation method
- (B) SiO_2 is removed by smelting process with CaO in the form of slag
- (C) Tin oxide is reduced by carbon reduction method to tin metal
- (D) Tin is refined by Hall Herault method
4. Which of the following is/are true for the complex compound with IUPAC name as :
Dichloridobis(ethane-1, 2-diamine)cobalt(III) sulphate
- (A) Its formula is $\left[\text{Co} \left(\begin{array}{c} \text{C H}_2 - \text{NH}_2 \\ | \\ \text{CH}_2 - \text{NH}_2 \end{array} \right)_2 \text{Cl}_2 \right] \text{SO}_4$
- (B) The co-ordination number of central metal ion is 6
- (C) The effective atomic number of central metal is 32
- (D) The compound can be categorised as a heteroleptic complex

Space for Rough Work

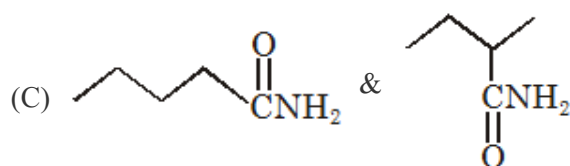
5. Which of the following is/are correctly matched :



Positional isomers



Tautomers

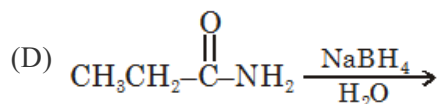
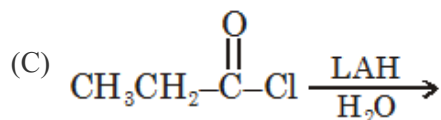
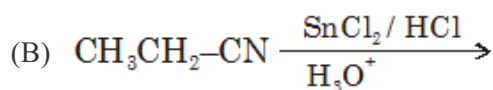
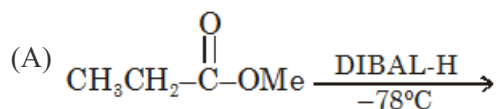


Chain isomers



Metamers

6. Which of the following reaction gives aldehyde as the product :



Space for Rough Work

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Paragraph for Questions 7 and 8

Polytropic process obeys the relation $PV^x = \text{constant}$ where x is a polytropic index. Value of x decides the type of process that can be obtained.

7. A process follows the equation $PV^3 = K$. If 1 mole $\text{CO}_2(\text{g})$ compressed from 1 atm to 27 atm from initial temperature of 300 K.

Given $C_v = \frac{5}{2}R$, $R = 0.08 \text{ litre atm / K mole}$, $R = 2 \text{ cal/ K mole}$

Incorrect option is :

- | | |
|--|---|
| (A) Final volume reached is 8 litre | (B) Final temperature will be 2700 K |
| (C) Work done during process is 96 lit atm | (D) Work done during process is 96 cal. |

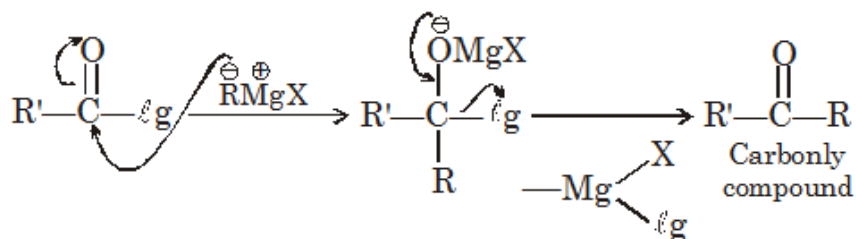
8. 1 mole ideal monoatomic gas follows process $\frac{T}{V^2} = \text{constant}$, then molar heat capacity of gas is (in terms of R)

- | | | | |
|---------|----------|----------|-----------|
| (A) R | (B) $2R$ | (C) $3R$ | (D) $R/2$ |
|---------|----------|----------|-----------|

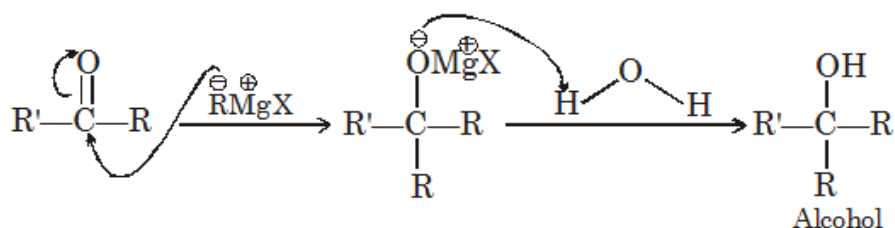
Space for Rough Work

Paragraph for Questions 9 and 10

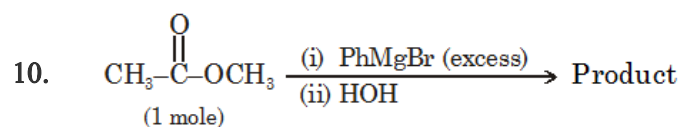
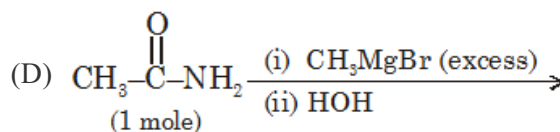
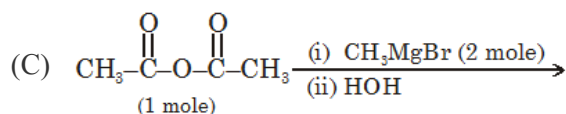
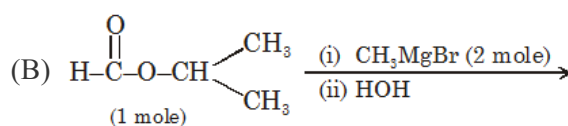
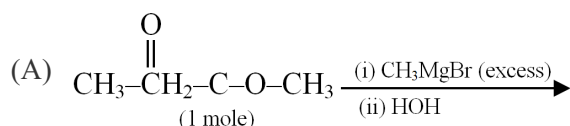
Generally acid derivatives gives SN^{AE} type of reaction with Grignard reagent.



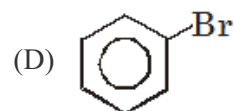
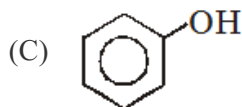
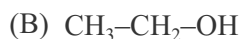
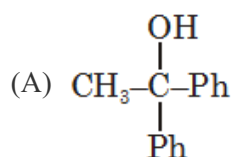
Carbonyl compound further react with Grignard reagent and give alcohol as product by Nucleophilic addition reaction.



9. Which of the following reaction gives only 2° alcohol as organic product :



Product can be :



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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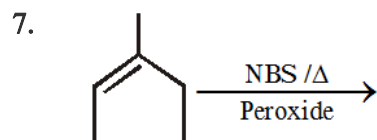
1. For the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$,
Calculate $\Delta_r S^\circ$ (in J/mol-K) at 300K.
Given : $(\Delta H_f^\circ)_{\text{NH}_3, 300\text{K}} = -46\text{kJ/mol}$
 $(\Delta G_f^\circ)_{\text{NH}_3, 300\text{K}} = -16\text{kJ/mol}$
2. Enthalpy of hydrogenation of ethylene is -125 kJ /mole. If bond enthalpy of C-H, C-C and H-H bonds are 414, 347 and 435 kJ/mole respectively. Calculate bond enthalpy of C=C. (in kJ/mole)
3. When a beam of photon having wavelength 155nm falls on a metal which has a work function of 5.6 eV, the maximum KE of the photoelectrons (in eV) possible is..[Given : $hc = 1240 \text{ eV} \cdot \text{nm}$]
4. One of the following is not an ore of iron. Give the atomic number of element extracted from it.
(i) Haematite (ii) Limonite
(iii) Siderite (iv) Malachite

Space for Rough Work

5. During extraction of iron, CO is the chief reducing agent. If 72 Kg of carbon is available and produced CO is completely used to reduce Haematite (present in excess amount) then calculate how much Kg of Iron is obtained after reduction.

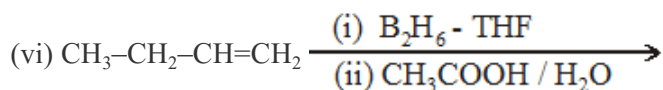
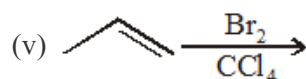
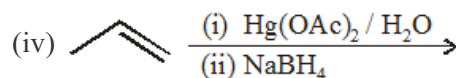
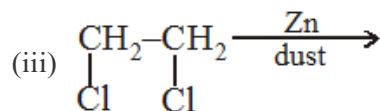
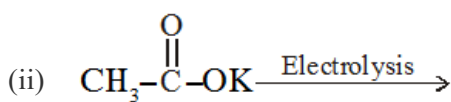
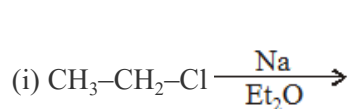
(Given : atomic weight of C-12 g/mol, O-16 g/mol, Fe-55.85 g/mol)

6. If the value of spin quantum number for the electron in the outermost shell of Cr atom is considered to be $+\frac{1}{2}$. Then find the sum of all quantum number for outermost shell electron.



Total monobrominated product (Possible) :

8. Number of following reactions which produces hydrocarbon as major product ?



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}, f(x) = \frac{\ln(x^2 + e^{2x})}{\ln(x^6 + e^x)}$. If $p = \lim_{x \rightarrow 0} f(x)$, $q = \lim_{x \rightarrow \infty} f(x)$, $r = \lim_{x \rightarrow -\infty} f(x)$, then
 (A) $p = q$ (B) $q = r$ (C) $p + \frac{1}{r} = 5$ (D) $q + \frac{1}{r} = \frac{7}{3}$
2. Area bounded by $y = \frac{4}{1+x^2}$, x-axis is bisected by the line $y = c$, then
 (A) $c > 2$ (B) $0 < c < 2$ (C) $0 < c < 1$ (D) $c < 4$
3. Consider the family of curves $y = Ae^x + Be^{2x} + Ce^{3x}$ where A, B, C are arbitrary constants. Then the differential equation of this family
 (A) has degree 1 (B) has order 3
 (C) is $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$ (D) is $\left(\frac{d^3y}{dx^3}\right)^2 = \frac{d^2y}{dx^2} + \frac{dy}{dx}$

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4. $\int \frac{x^2}{(x \sin x + \cos x)^2} dx$ is equal to (C is constant of integration)

(A) $\frac{\sin x - x \cos x}{x \sin x + \cos x} + C$

(B) $\frac{\sin x + x \cos x}{x \sin x + \cos x} + C$

(C) $\tan(x + \tan^{-1}x) + C$

(D) $\tan(x - \tan^{-1}x) + C$

5. Given that $\lim_{t \rightarrow 0} \frac{f(a+t) - f(a-t)}{t} = 10$, (given $a \in \mathbb{R}$) then identify the **INCORRECT** statement(s)

(A) $f'(a) = 5$

(B) $f(x)$ is continuous at $x = a$

(C) $\lim_{x \rightarrow a} f(x)$ exists

(D) $f(a)$ is finite

6. $P = \lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} \sin(\sqrt{t}) dt}{\tan x - x},$

$Q = \lim_{x \rightarrow 0^-} \frac{\int_0^{x^2} \sin(\sqrt{t}) dt}{\tan x - x},$

$L = \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin(\sqrt{t}) dt}{\tan x - x}.$ Then

(A) $P = 2$

(B) $Q = 2$

(C) L does not exist

(D) $|P| + |Q| = 2L$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $f : \mathbb{R} \rightarrow (0, \infty)$, $f(x)$ be a differentiable function satisfying $f(x+y) \cdot f(x-y) = f^2(x) \forall x, y \in \mathbb{R}$.

Further $f(0) = 1$, $f'(0) = \frac{1}{\log_2 e}$

7. $f(x)$ is
- | | |
|---------------------------|------------------------------|
| (A) One-one but not onto | (B) Onto but not one-one |
| (C) One-one and onto both | (D) Neither one-one nor onto |
8. $\int_0^2 f^{-1}(x) dx$ is equal to
- | | | | |
|---------------|------------------|---|--|
| (A) $3 \ln 2$ | (B) $3 \log_2 e$ | (C) $\log_2 \left(\frac{4}{e^2} \right)$ | (D) $\ln \left(\frac{4}{e^2} \right)$ |
|---------------|------------------|---|--|

Paragraph for Questions 9 and 10

$f(x) = \sin^{-1}(\sin 2) + \sin^{-1}(\sin 3) + \sin^{-1}(\sin x)$, $x \in \mathbb{R}$. Let $x_1, x_2, x_3 \in \mathbb{Z}$ such that $f(x_1), f(x_2), f(x_3)$ are all integers. Then

9. $f(x_1) + f(x_2) + f(x_3)$ is equal to
- | | | | |
|-------|--------|--------|--------|
| (A) 3 | (B) 15 | (C) 18 | (D) 20 |
|-------|--------|--------|--------|
10. $\sum_{r=1}^{10} f'(r)$ is equal to
- | | | | |
|--------|-------|-------|--------|
| (A) 10 | (B) 0 | (C) 2 | (D) -2 |
|--------|-------|-------|--------|

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SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Number of tangents drawn to the curve $y = e^{-\{x\}}$ which pass through $\left(-\frac{1}{2}, 0\right)$ is $(\{x\})$ denotes fractional part of x
2. Let $A = \sum_{r=1}^9 \cot^{-1} \left(\frac{r-5}{r^2 - 10r + 50} \right)$ and $B = \int_1^9 \cot^{-1} \left(\frac{x-5}{x^2 - 10x + 50} \right) dx$, then determine the value of $\frac{A}{B}$
3. $f : \mathbb{R} \rightarrow (0, \infty)$, $f(x)$ is a twice differentiable function such that $f'(x) = xf(x)$ for $x = 1, 2, 3, 4$, then find minimum number of roots of the equation $f(x) \cdot f''(x) = (f'(x))^2 + (f(x))^2$
4. P is a point outside a given circle C_1 . Line through point P meets C_1 at Q and R, this line also passes through centre O of the circle C_1 . Chord of contact of tangents drawn to C_1 from P is bisected at M. If $PQ = 6$, $PM = 8$, find radius of circle C_1

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5. $\int_1^2 \frac{6x^2 + 5x + 4}{x^3 + x^2 + x} dx = \ln(k)$, then find the value of k
6. $\int e^x (x^3 + 4x^2 + 3x + 1) dx = e^x \cdot f(x) + C$, where C is constant of integration. Then determine the value of $f(2)$
7. Let V_1 be maximum possible volume of a right circular cylinder inscribed in a given sphere of volume V_2 . Then find the value of $\left(\frac{V_2}{V_1}\right)^2$
8. A circle touches $y = x^3 + x^2 + x + 1$ at $A(1, 4)$ and passes through origin. If equation of this circle is $x^2 + y^2 + 2gx + 2fy = 0$, determine the value of g
-

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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